

Reasoning about derived classes:

derived-defaultCtor.cpp

```

1 class Sphere {
2     public:
3         Sphere(double d) { /* ... */ }
4 };
5
6 class Ball : Sphere() {
7
8
9 }
10
11 int main() {
12     Ball b;
13 }

```

Abstract Class:

1. [Requirement]:
2. [Syntax]:
3. [As a result]:

Reasoning about virtual classes:

virtual-ctor.cpp

```

15 class Sphere {
16     public:
17         virtual Sphere();
18 };
19
20 class Ball : Sphere() {
21     public:
22         _____;
23 }
24

```

virtual-dtor.cpp

```

15 class Sphere {
16     public:
17         virtual ~Sphere();
18 };
19
20 class Ball : Sphere() {
21     public:
22         _____;
23 };
24

```

In what order does the derived classes ctor/dtor get called?

Assignment Operator

I glossed over a VERY important detail of the assignment operator:

Consider the following program:

assignmentOpSelf.cpp

```

1 #include "Sphere.h"
2
3 int main() {
4     cs225::Sphere s(10);
5     s = s;
6     return 0;
7 }

```

Sphere.cpp

```

10 void Sphere::_destroy() { delete[] props_; }
11
12 void Sphere::_copy(const Sphere &other) {
13     r_ = other.r_;
14     props_max_ = other.props_max_;
15     props_ct_ = other.props_ct_;
16     props_ = new std::string[10];
17     for (unsigned i = 0; i < props_ct_; i++) {
18         props_[i] = other.props_[i];
19     }
20 }
21
22 Sphere& Sphere::operator=(const Sphere &other) {
23
24
25
26     _destroy();
27     _copy(other);
28     return *this;
29 }

```

Big Idea: Abstract Data Type

List ADT

Purpose	Function Definition

What types of List do we want?

Solution:

Function templates:

functionTemplate1.cpp	
1	
2	
3	T maximum(T a, T b) {
4	T result;
5	result = (a > b) ? a : b;
6	return result;
7	}

functionTemplate2.cpp

```
1
2
3 T maximum(T a, U b) {
4   return (a > b) ? a : b;
5 }
```

Class templates:

List.h

```
1 #ifndef LIST_H
2 #define LIST_H
3
4
5
6 class List {
7   public:
8
9
10
11   private:
12
13
14 };
15
16 #endif
```

List.cpp

```
1
2
3
4
5
6
7
8
9
10
11
12
```

CS 225 – Things To Be Doing:

1. Lab2 is due March 8
2. MP2 is due March 16
3. Max Extra Credit w/ Part 1 by March 9